A. Multiple-choice questions

|!|BCEMB0380000002|!| ★

Which of the following statements about viruses is/are correct?

(1) The nucleic acid of a virus is surrounded by a nuclear membrane.
(2) Most viruses are specific to certain types of host cells.
(3) Viruses multiply inside host cells by cell division.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0380000004|!| ★

Which of the following is/are microorganisms that lack a true nucleus?

(1) bacteria
(2) viruses
(3) yeasts

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0380000010|!| ★

Which of the following statements correctly compare *Escherichia coli* and *Paramecium*?

(1) Both are unicellular.
(2) Both lack a true nucleus.
(3) Both lack chloroplasts.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

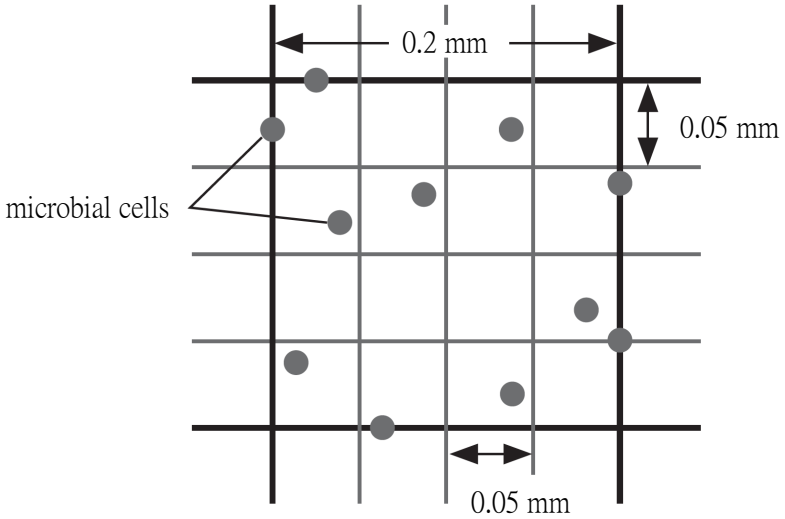
□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0380000019|!| ★

A student used a cell counter to measure the number of cells in a microbial sample. The diagram below shows the distribution of microbial cells within a large square of 0.2 mm × 0.2 mm of the counting chamber. The counting chamber is 0.1 mm deep.



Which of the following correctly shows the number of microbial cells in 1 cm^3 of the sample?

A. 2×10^3
B. 2×10^6
C. 2.75×10^3
D. 2.75×10^6

□

##B##

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Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0380000020|!| ★

0.1 cm^3 of a bacterial sample with five times of 10-fold dilution was transferred to a nutrient agar plate. After incubation, 165 colonies can be observed on the plate. What is the number of bacterial cells per cm^3 of the original sample?

A. 1.65×10^6 cells per cm^3
B. 1.65×10^7 cells per cm^3
C. 1.65×10^8 cells per cm^3
D. 1.65×10^9 cells per cm^3

□

##C##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMC0390000001|!|

Which of the following are the advantages of using microorganisms for producing commercial products?

(1) They can be grown in large numbers in a relatively small space.
(2) Their genetic materials can be easily modified.
(3) The cost for maintaining the microbial culture is cheap.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0400000014|!| ★

Directions: The following two questions refer to the table below, which shows the features of four types of bacteria that can cause food poisoning.

Bacterium	Temperature range for growth (°C)	Production of heat-resistant toxins
W	-1.5 – 37	✓
X	25 – 106	×
Y	4 – 50	✓
Z	10 – 72	×

Cooking the food thoroughly can prevent food poisoning caused by

A. Z only.
B. X and Z only.
C. Y and Z only.
D. W, Y and Z only.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0400000015|!| ★

Which of the following food preservation methods can kill bacterium Y?

(1) freezing
(2) pasteurization
(3) autoclaving

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0400000018|!|

Directions: The following two questions refer to the table below, which shows part of the procedures of two methods of microbial control (X and Y) used in the treatment of milk.

Method	Procedure
X	Heat the milk to 138 °C for 2 – 4 seconds. The milk is then quickly cooled.
Y	Heat the milk to 72 °C for 15 seconds. The milk is then quickly cooled.

Which of the following combinations correctly shows methods X and Y?

	X	***Y***
A.	autoclaving	pasteurization
B.	pasteurization	ultra high temperature (UHT) treatment
C.	ultra high temperature (UHT) treatment	pasteurization
D.	ultra high temperature (UHT) treatment	autoclaving

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0400000019|!| ★

Which of the following statements is/are correct?

(1) Both methods can extend the shelf life of milk.
(2) Both methods can sterilize the milk.
(3) Milk treated with method Y does not need refrigeration.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

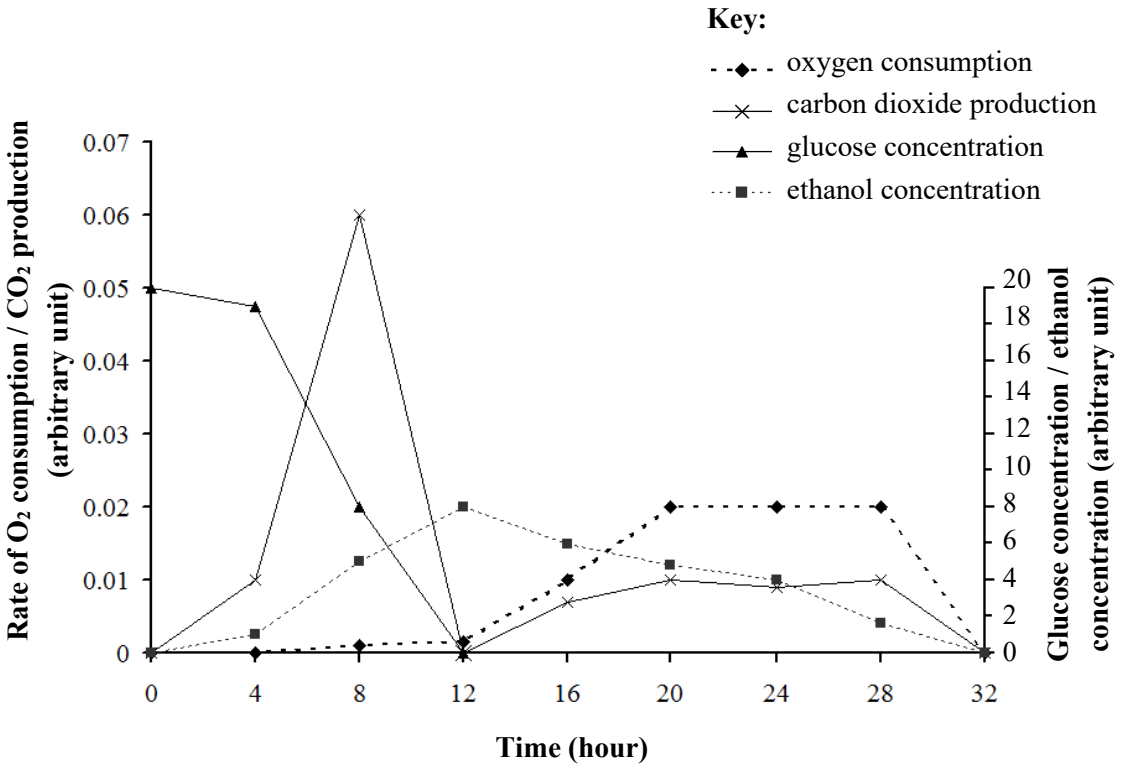
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B. Structured questions

|!|BCELA0380000001|!| ★★ **Cross topic**

The baker's yeast (*Saccharomyces cerevisiae*) carries out alcoholic fermentation. In this process, glucose is broken down in the absence of oxygen to give ethanol. When glucose is insufficient, the baker's yeast converts ethanol to acetyl coenzyme A.

The graph below shows in a batch of yeast culture the changes in the rate of oxygen consumption, the rate of carbon dioxide production, glucose concentration and ethanol concentration over 32 hours.



(a) Suggest what happened in the first 12 hours that caused the changes in the four parameters shown. (4 marks)

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(b) Explain for the changes in the three parameters after the 12th hour. (2 marks)

(c) Describe ***one*** method of determining the population size of yeast in a liquid culture. (4 marks)

##

(a) At start, the glucose concentration is high. (1)
Yeasts respire anaerobically to break down glucose and produce ethanol. (1)
Hence, glucose concentration decreased, carbon dioxide production and ethanol concentration increased, while oxygen consumption remained low. (1)
At the 12th hour, all the glucose were used up, so respiration stopped and carbon dioxide production stopped. (1)

(b) In the absence of glucose, yeasts convert ethanol to acetyl coenzyme A which enters the Krebs cycle and later in oxidative phosphorylation. (1)
In this way, yeasts consume ethanol and oxygen, and produce ATP and carbon dioxide. (1)

(c) Conduct serial dilution on the culture. (1)
Transfer a fixed amount of the diluted sample to a nutrient agar plate, and spread the diluted sample evenly over the surface of the plate. (1)
Incubate the plate until a considerable number of colonies formed. (1)
Count the number of colonies and calculate the number of viable cells using the dilution factor. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	2
(b)	Applying concepts	2
(c)	Understanding basic concepts	4

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|!|BCELB0380000007|!| ★

Different kinds of microorganisms have distinct characteristics.

(a) State the characteristics of the following kinds of microorganisms:

(i) halophiles (salt-loving microorganisms) (1 mark)

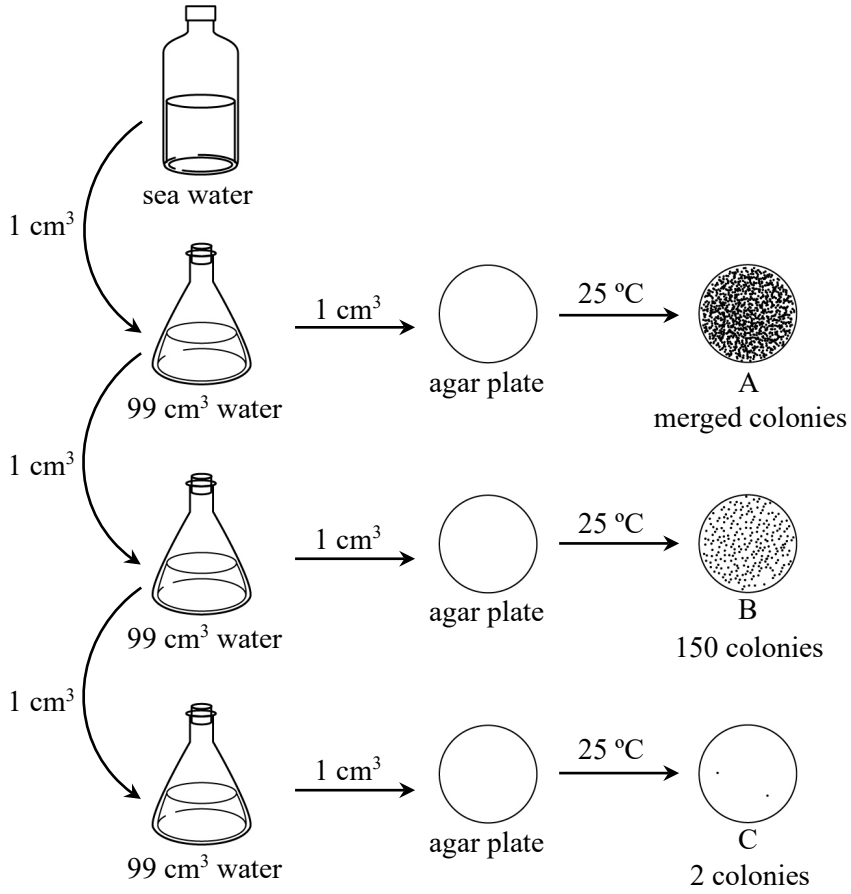
(ii) facultative anaerobes (1 mark)

(b) (i) State ***three*** aseptic techniques that are used to prepare nutrient agar plates and transfer bacteria from nutrient broths to the plates. (3 marks)

(ii) After inoculation, the Petri dish is sealed with adhesive tape and incubated at 25ºC. Explain why the plates should not be sealed completely and should be placed upside down during incubation. (2 marks)

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(c) The following diagram shows a method that is used to calculate the number of bacteria in a sea water sample.



(i) Explain why the result from plate B should be used instead of those from plates A and C to calculate the number of bacteria in the sea water sample. (2 marks)

(ii) Using the data shown in the diagram, calculate the number of bacterial cells per cubic centimetre of sea water. Show your working. (2 marks)

##

(a) (i) They require high salt concentrations to grow. (1)

(ii) They grow better with oxygen but can survive under anaerobic conditions. (1)

(b) (i) Any ***three*** of the following: (3)

- disinfect the bench / hands with 70% ethanol
- sterilize the culture medium with the use of an autoclave
- sterilize the inoculating loop by flaming it to red hot over a Bunsen burner before inoculation
- flame the mouth of the container that contains the bacterial culture before and after inoculation
- lift the lid of the Petri dish containing the nutrient agar as little as possible when transferring the bacteria

(Or other reasonable answers)

(ii) If the agar plates are sealed completely, the growth of aerobic bacteria will be inhibited due to a lack of oxygen. (1)

If the plates are not placed upside down, water condensed on the lids will drop onto the bacterial culture. (1)

(c) (i) Counting the number of colonies on Plate B would produce results with the least statistical error. (1)

Counting the number of colonies on Plate C would produce statistically invalid results while counting the number of colonies on Plate A would be difficult and inaccurate. (1)

(ii) Dilution factor = 10^4 (1)

Number of bacterial cells per cm^3 sea water = $150 \times 10^4 = 1\ 500\ 000$ cells (1)

(11 marks)

##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	3
(b)(ii)	Practical skills	2
(c)(i)	Analysing data	2
(c)(ii)	Mathematical skills	2

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|!|BCELB0390000004|!| ★

The forestomachs of baleen whales have been found to contain more than 1000 different species of anaerobic bacteria. Several species have been identified that can break down hydrocarbons, which are components of crude oil.

(a) Some scientists think that changes in the habitats of baleen whales led to the evolution of these species of bacteria that can break down hydrocarbons.
What is the possible change in the habitats of baleen whales? (1 mark)

(b) Explain why the bacteria isolated from the forestomachs of baleen whales can be used to break down a thick layer of oil spill more effectively. (3 marks)

(c) Briefly describe how bacteria that can break down oil can be produced on a large scale by genetic engineering. (3 marks)

(d) State ***two*** environmental hazards caused by the bacteria produced in (c). (2 marks)

##

(a) Their habitats are polluted by crude oil. (1)

(b) A thick layer of oil spill means a thicker barrier to oxygen. (1)
Bacteria from baleen whales are anaerobes, (1)
thus can break down oil in deeper layer much faster. (1)

(c) The gene that codes for enzymes that can break down hydrocarbons is isolated from the bacteria that can break down hydrocarbons. (1)
The gene is then introduced into another bacterial cell that is easier to culture. (1)
The GM bacteria can then be cultured on a large scale in a fermenter. (1)

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(d) GM bacteria may out-compete wild types and upset the ecological balance. (1)
GM bacteria may transfer genes that are unfavourable to other organisms to the wild types. (1)
(Or other reasonable answers)

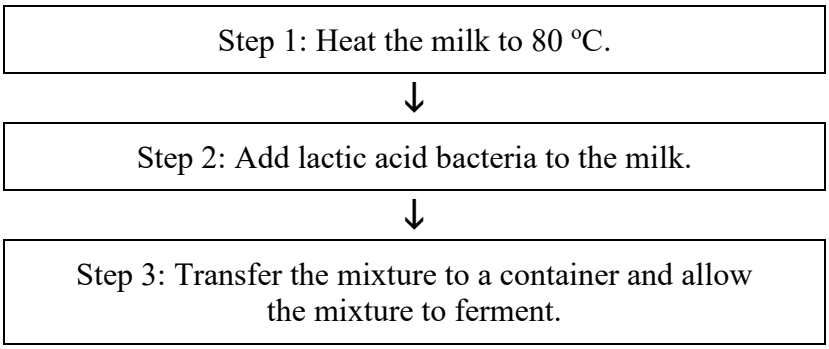
(9 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Understanding basic concepts	3
(d)	Critical thinking	2

|!|BCELB0400000021|!| ★ **Cross topic**

Yoghurt is a food produced by bacterial fermentation of milk. The flow chart below outlines the process of making yoghurt.



(a) Explain why the milk has to be heated to 80 ºC in step 1. (2 marks)

(b) In step 2, the milk has to be cooled to 40 ºC before adding lactic acid bacteria. Explain why. (1 mark)

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(c) Suggest why yoghurt should be stored in the refrigerator before consumption. (2 marks)

(d) Although yoghurt is made from milk, the texture of yoghurt is different from that of milk. And also the shelf life of yoghurt is longer than that of milk.

(i) Describe and explain the change in the viscosity when milk turns into yoghurt. (3 marks)

(ii) Explain why yoghurt spoils later than milk when both of them are stored in the same condition. (2 marks)

##

(a) To kill the microorganisms which may convert the milk to unwanted products (1)
To remove oxygen in the milk to provide an anaerobic environment for the microorganisms to carry out fermentation (1)

(b) To prevent the lactic acid bacteria from being killed by the high temperature. (1)

(c) Low temperature slows down the growth of microorganisms (1)
to prevent the spoilage of yoghurt. (1)

(d) (i) Lactic acid bacteria carry out lactic acid fermentation. (1)
The lactic acid produced lowers the pH and causes the proteins in milk to denature and coagulate. (1)
As a result, the viscosity of the fermented milk increases. (1)

(ii) The pH of yoghurt is lower than that of milk (1)
which inhibit the growth of spoilage microorganisms. (1)

(10 marks)

##

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	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	1
(c)	Applying concepts	2
(d)(i)	Applying concepts	3
(d)(ii)	Applying concepts	2

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